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Deluge Scripting in Zoho Creator

1. Aim

To learn the Deluge scripting language of Zoho Creator by writing and running ten standard stand-alone programs that use variables, conditions, loops, lists, maps, string functions and date functions.

2. Introduction

2.1 Deluge

Deluge (Data Enriched Language for the Universal Expression) is the scripting language built into Zoho Creator. It adds logic to an application without a separate programming environment. A Deluge script is written in the script editor of a workflow, a function, a button, a schedule or an approval. The result of a script can be shown with the info statement, which is useful while learning and testing.

2.2 Where Deluge Scripts Are Used

Table 2 lists the places in a Zoho Creator application where Deluge scripts run. The ten programs of this experiment are stand-alone programs that need no form or other setup and can be used in any of these places.

3. Objectives of the Study

- To declare variables and use arithmetic operators and the basic data types of Deluge.
- To control the flow of a script with if / else conditions and for each loops.
- To store data in lists and maps and to process text and dates with built-in functions.
- To solve common problems such as finding the largest number, counting even and odd numbers and calculating a factorial.

4. Procedure

1. Open Zoho Creator and open a practice application in Edit mode, then click Workflow in the top bar.
2. Open the Functions section and click + New Function to create a standalone function for each program, for example program_01_variables.
3. Type or paste the Deluge script of the program in the script editor.
4. Click Save, then click Execute (Run) to run the function.
5. Read the output of the info statements in the output window of the editor.
6. Take a screenshot of the editor with the output for every program.

5. Detailed Analysis

Table 1 lists the ten programs and the Deluge features used by each.

No.	Program	Deluge features used
1	Variables and arithmetic operators	Variables, arithmetic, %
2	if / else if condition	if, else if, else
3	for each loop: sum and average	List, for each, size
4	List operations	add, get, contains, sort, removeElement
5	Map operations	Map, put, get, containKey, remove, keys
6	String functions	length, toUpperCase, subString, indexOf, replaceAll, toList
7	Date functions	zoho.currentdate, addDay, addMonth, toString
8	Largest and smallest in a list	List, for each, if, comparison
9	Counting even and odd numbers	for each, if / else, % operator
10	Multiplication table and factorial	for each, info, multiplication

Table 1: Deluge programs written in this experiment

Table 2 shows where Deluge scripts are used in a Zoho Creator application.

Place	Use of the Deluge script
Form workflow	Fills fields, validates data and updates other records when a record is created or edited
Button	Runs a script when the user clicks a button on a form, report or page
Function	A reusable script that is called from workflows, buttons and schedules
Schedule	Runs a script at a fixed date and time
Approval	Runs a script when a record is approved or rejected
Blueprint	Runs a script when a record moves from one stage to the next

Table 2: Places where Deluge scripts are used

The ten programs follow, each with its script and the output it gives.

Deluge script	Sample output

`price = 250;`
`quantity = 4;`
`discount = 10;`
`subtotal = price * quantity; discountAmt`
`= subtotal * discount / 100; total =`
`subtotal - discountAmt;`
`info "Subtotal : " + subtotal;`
`info "Discount : " + discountAmt;`
`info "Total : " + total;`
`info "Remainder : " + (subtotal % 3);`
	`Subtotal : 1000`
	`Discount : 100.0`
	`Total : 900.0`
	`Remainder : 1`

Program 1: Variables and arithmetic operators

Deluge script	Sample output

marks = 72;
if(marks >= 90)
{
grade = "A+";
}
else if(marks >= 75)
{
grade = "A";
}
else if(marks >= 60)
{
grade = "B";
}
else
{
grade = "C";
}
info "Marks : " + marks;
info "Grade : " + grade;
	Marks : 72
	Grade : B

Program 2: if / else if condition

Deluge script	Sample output

marks = List();
marks.add(78);
marks.add(85);
marks.add(92);
marks.add(66);
sum = 0;
for each m in marks
{
sum = sum + m;
}
average = sum / marks.size();
info "Sum : " + sum; info
"Average : " + average;
	Sum : 321
	Average : 80.25

Program 3: for each loop: sum and average

Deluge script	Sample output

names = List();
names.add("Ravi");
names.add("Anita");
names.add("Meena");
info "Size : " + names.size();
info "Second : " + names.get(1);
info "Has Anita : " + names.contains("Anita");
info "Sorted : " + names.sort(true);
names.removeElement("Ravi");
info "After remove : " + names;
	Size : 3
	Second : Anita
	Has Anita : true
	Sorted : [Anita, Meena, Ravi]
	After remove : [Anita, Meena]

Program 4: List operations

Deluge script	Sample output

student = Map();
student.put("Name", "Anita");
student.put("Dept", "CSE");
student.put("Marks", 88);
info student.get("Name");
info student.containKey("Dept");
student.remove("Dept");
info student.keys();
info student.size();
	Anita
	true
	[Name, Marks]
	2

Program 5: Map operations

Deluge script	Sample output

text = "Zoho Creator Lab";
info text.length();
info text.toUpperCase(); info
text.toLowerCase(); info
text.subString(5, 12); info
text.indexOf("Creator");
info text.replaceAll("Lab", "Record");
info text.contains("Zoho");
words = text.toList(" ");
info words.size();
	16
	ZOHO CREATOR LAB
	zoho creator lab
	Creator
	5
	Zoho Creator Record
	true
	3

Program 6: String functions

Deluge script	Sample output

today = zoho.currentdate;
info "Today : " + today;
info "Next week : " + today.addDay(7); info
"Next month : " + today.addMonth(1);
info "Day name : " + today.toString("EEEE");
info "Formatted : " + today.toString("dd/MM/yyyy");
info "Year : " + today.getYear();
	Today : 18-Oct-2026
	Next week : 25-Oct-2026
	Next month : 18-Nov-2026
	Day name : Sunday
	Formatted : 18/10/2026
	Year : 2026

Program 7: Date functions

Deluge script	Sample output

nums = List();
nums.add(45);
nums.add(12);
nums.add(78);
nums.add(33);
largest = nums.get(0);
smallest = nums.get(0);
for each n in nums
{
  if(n > largest)
  {
  largest = n;
  }
  if(n < smallest)
  {
  smallest = n;
  }
}
info "Largest : " + largest; info
"Smallest : " + smallest;
	Largest : 78
	Smallest : 12

Program 8: Largest and smallest in a list

Deluge script	Sample output

`nums = List();`
`nums.add(3);`
`nums.add(8);`
`nums.add(15);`
`nums.add(20);`
`nums.add(7);`
`evenCount = 0;`
`oddCount = 0;`
`for each n in nums`
`{`
`  if(n % 2 == 0)`
`  {`
`  evenCount = evenCount + 1;`
`  }`
`  else`
`  {`
`  oddCount = oddCount + 1;`
`  }`
`}`
`info "Even numbers : " + evenCount;`
`info "Odd numbers : " + oddCount;`
	`Even numbers : 2`
	`Odd numbers : 3`

Program 9: Counting even and odd numbers

Deluge script	Sample output

`nums = List();`
`nums.add(1);`
`nums.add(2);`
`nums.add(3);`
`nums.add(4);`
`nums.add(5);`
`fact = 1;`
`for each n in nums`
`{`
`  fact = fact * n;`
`  info "3 x " + n + " = " + (3 * n);`
`}`
`info "Factorial of 5 : " + fact;`
	`3 x 1 = 3`
	`3 x 2 = 6`
	`3 x 3 = 9`
	`3 x 4 = 12`
	`3 x 5 = 15`
	`Factorial of 5 : 120`

Program 10: Multiplication table and factorial

7. Output

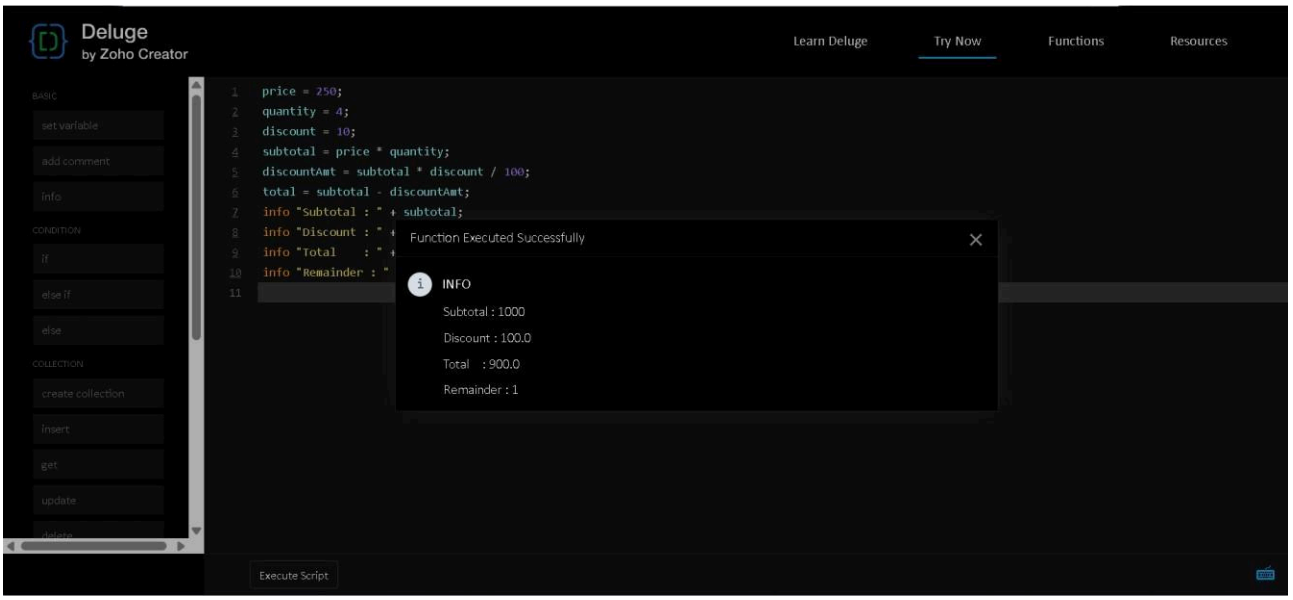


Figure 1: Program 1 (Variables and arithmetic operators) and its output

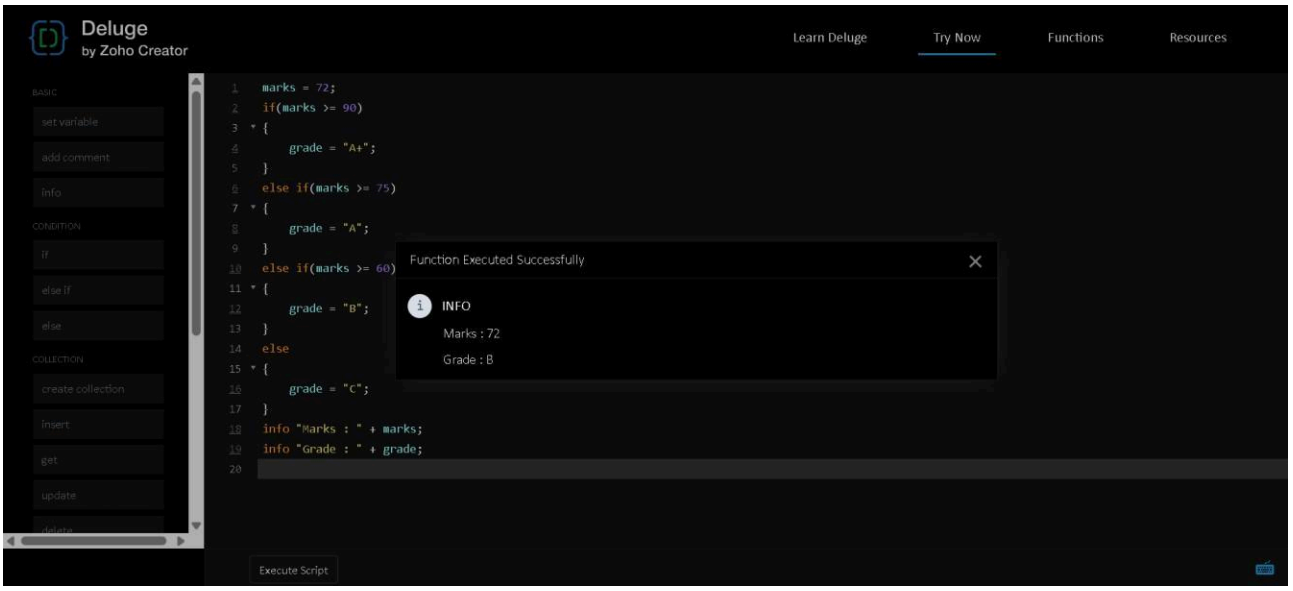


Figure 2: Program 2 (if / else if condition) and its output

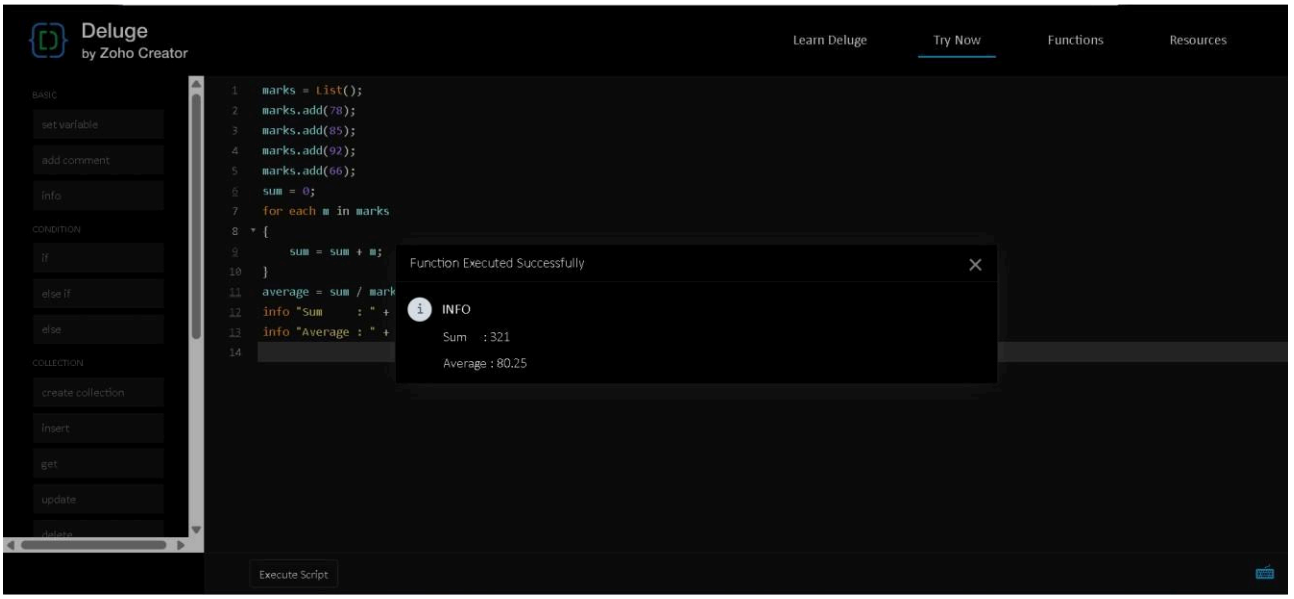


Figure 3: Program 3 (for each loop: sum and average) and its output

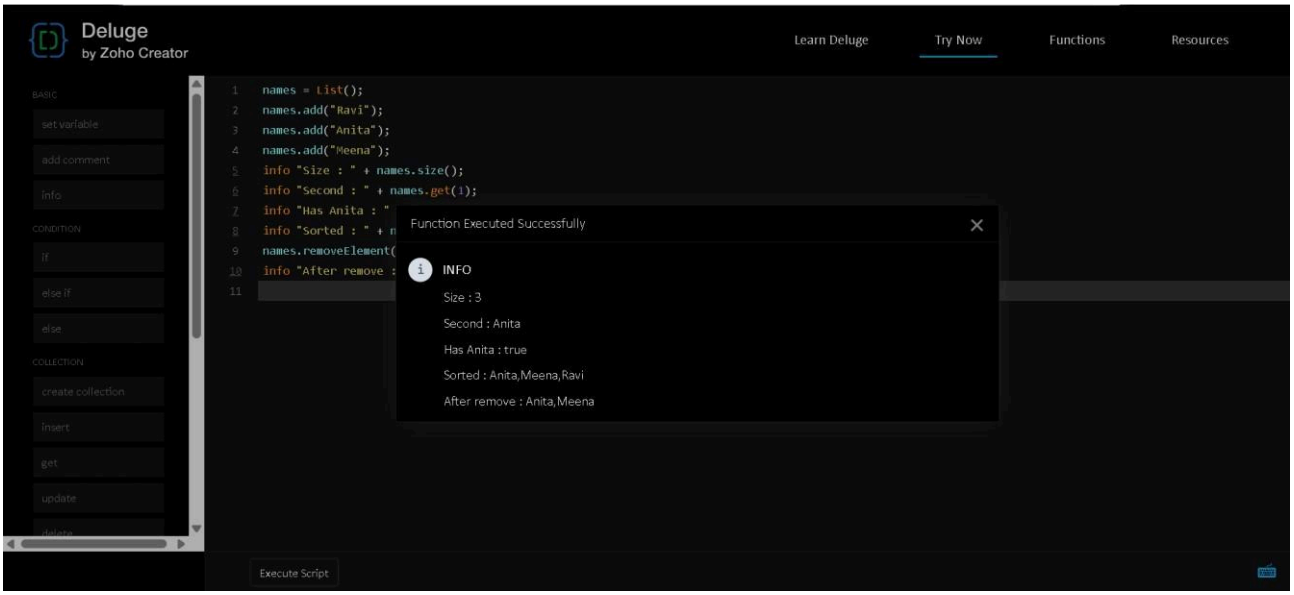


Figure 4: Program 4 (List operations) and its output

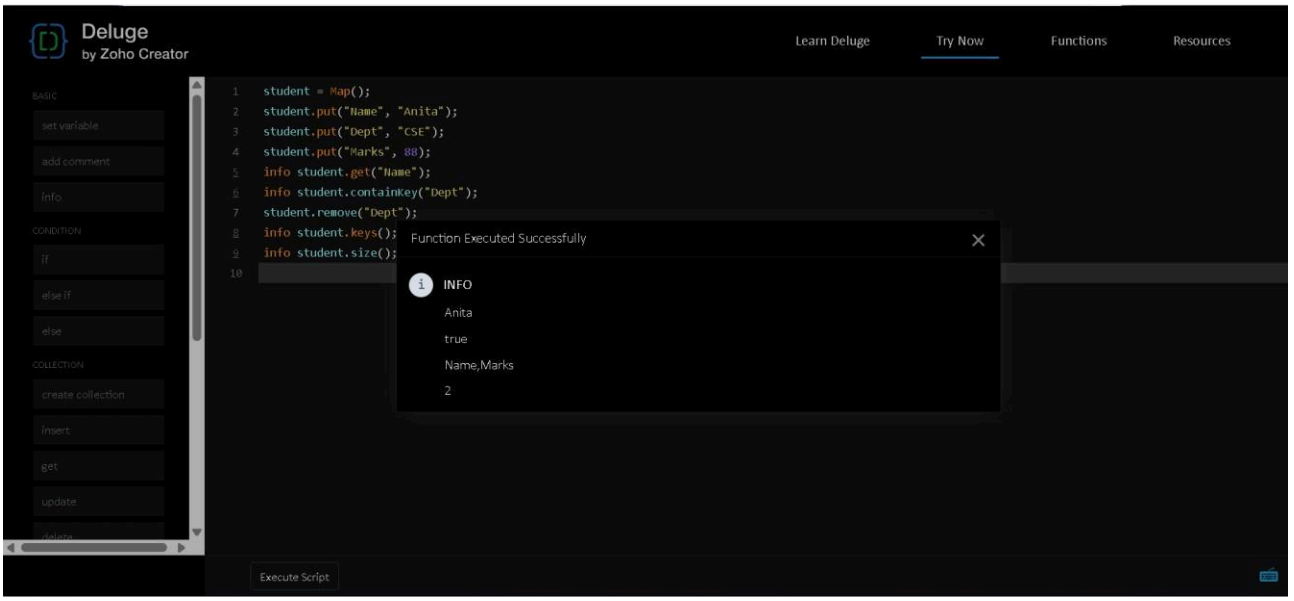


Figure 5: Program 5 (Map operations) and its output

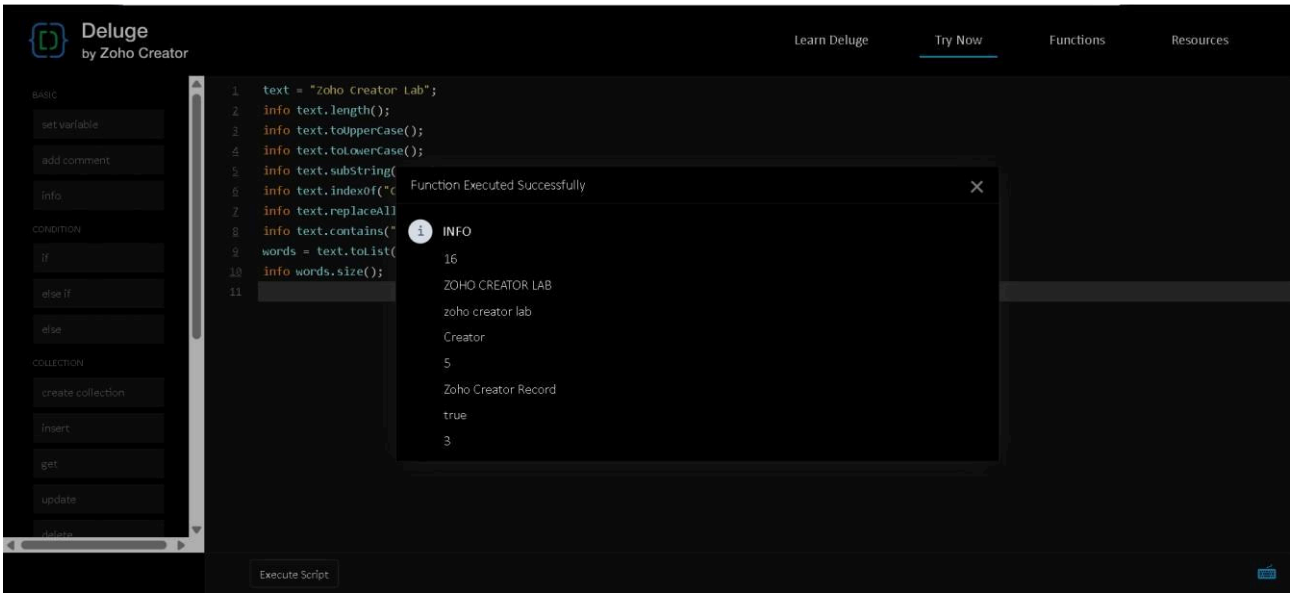


Figure 6: Program 6 (String functions) and its output

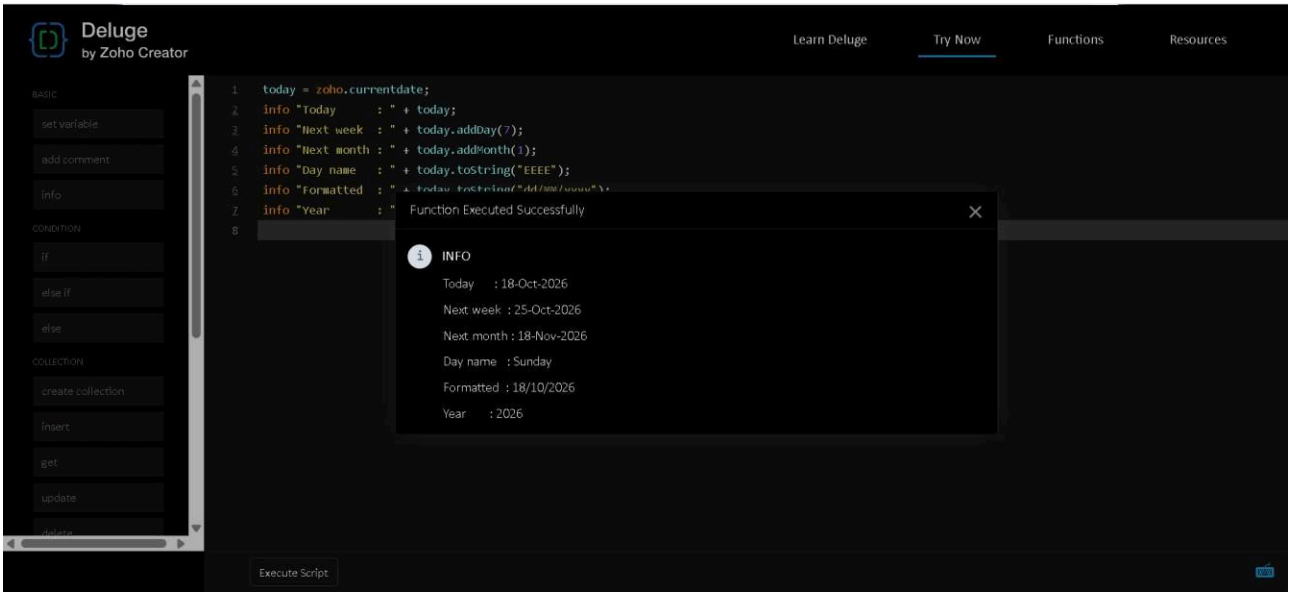


Figure 7: Program 7 (Date functions) and its output

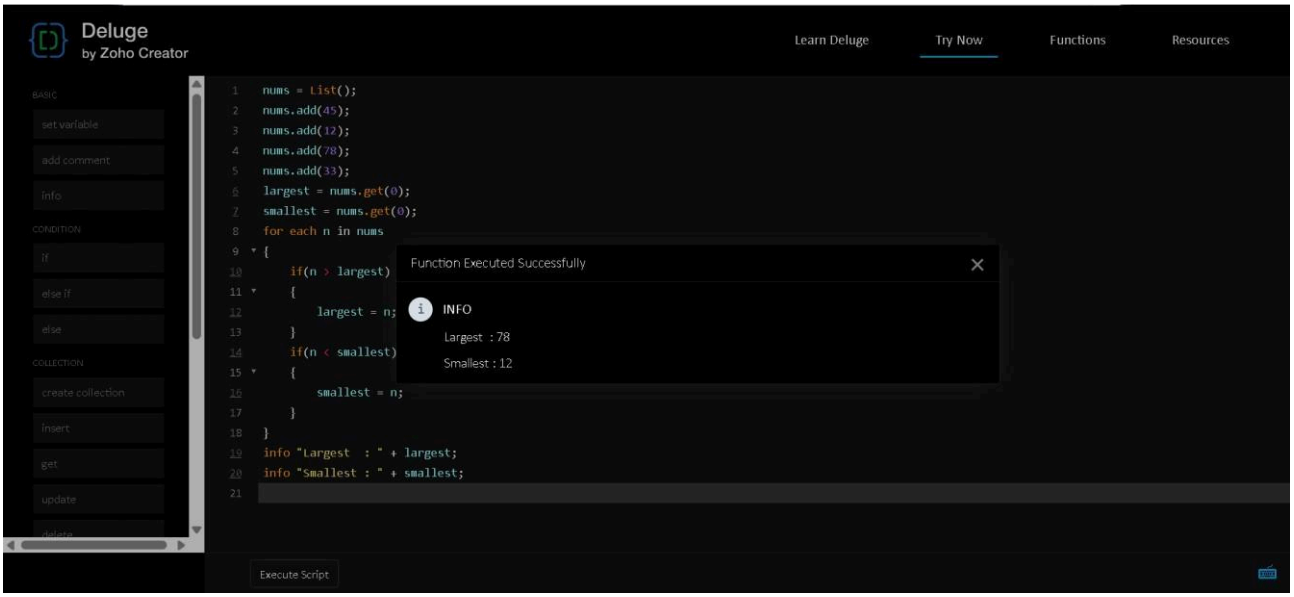


Figure 8: Program 8 (Largest and smallest in a list) and its output

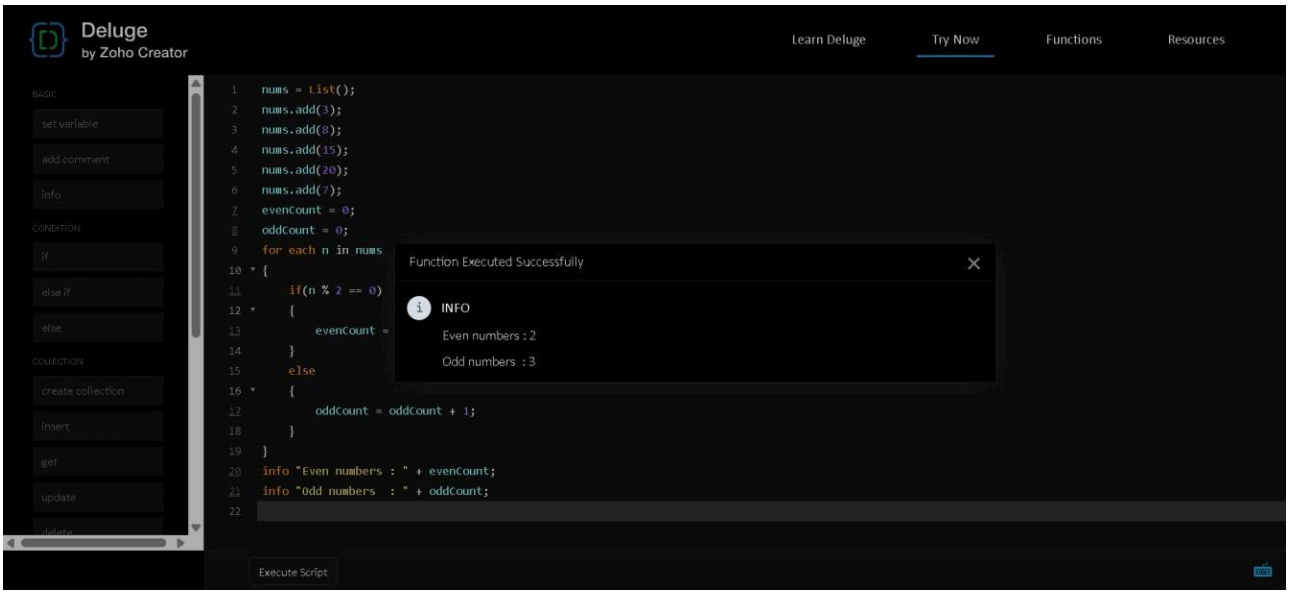


Figure 9: Program 9 (Counting even and odd numbers) and its output

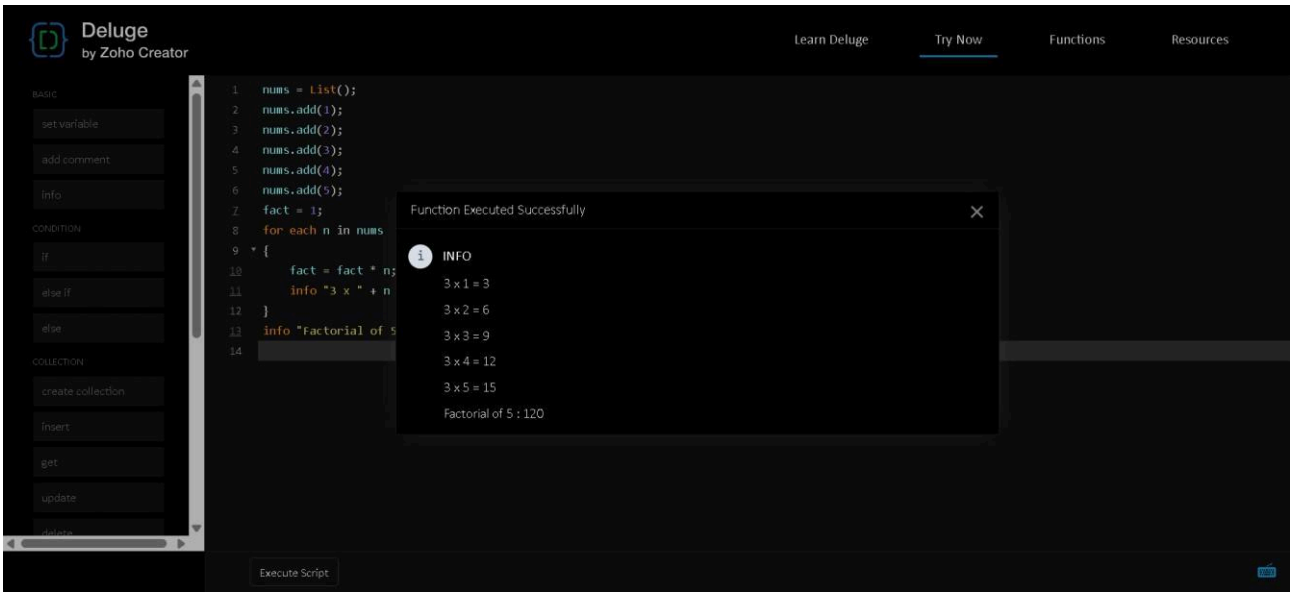


Figure 10: Program 10 (Multiplication table and factorial) and its output

7. Result

Ten standard Deluge programs were written and run in Zoho Creator. They covered variables, conditions, loops, lists, maps, string and date functions, and common number problems. Each script gave the expected output, which shows the basic building blocks used to add logic to any Zoho Creator application.